

WIC AI INNOVATOR CAPSTONE

Lumen

Defensible AML Decision Workbench

91.1%

Evidence Completeness

765

Tests Passing

9

Claim Types Locked

7

Control Layers

Leishka · Shivani · Ryan · Suli



The Problem

95%

of AML alerts are false positives

Source: LexisNexis Risk Solutions

45 min

manual data-gathering per case
(ACAMS benchmark)



Alert Fatigue

Analysts open 5 to 7 browser tabs per case just to find a customer ID, transaction history, and prior alerts.



AI Hallucinations Go Unchecked

Existing tools pass AI summaries directly to analysts. When the AI is wrong, no technical control catches it.



Rubber-Stamp Culture

Time pressure turns human review into a checkbox. Regulatory risk follows every unchallenged approval.



Our Answer

The model drafts. Code verifies. Humans decide.

01

Structured Claims First

The AI emits JSON claims from a closed 9-type vocabulary. No free prose. Every claim is a checkable data object.

02

Deterministic Verification Second

Python code checks every claim against source tables. Returns PASS, FAIL, or NEEDS REVIEW. No LLM judging another LLM.

03

Human Approval Last

Anti-rubber-stamp gate. A review is not complete until evidence, rationale, and a final action are recorded. The AI never decides.



Most AI compliance tools try to make analysts trust AI faster. Lumen does the opposite.

What We Actually Built

This separates us from teams who made a concept deck.



data/

Synthetic AML dataset (8 sources + audit)
30 customers, 187 transactions
Seeded demo cases and review records
Evidence completeness averages 91.1%



src/schema.py

Pydantic v2 validation models
Closed 9-type claim vocabulary
Single source of truth for all tables



src/llm_drafter.py

Anthropic tool-use API call
Forced structured claim output
3-gate schema enforcement
Never-raise fallback behavior



src/verifier.py

9 deterministic verifier functions
PASS / FAIL / NEEDS_REVIEW
Field-based and pattern-based checks



src/pipeline.py

Intended production wiring
Chains draft, verify, and gate
Demo calls the verifier directly



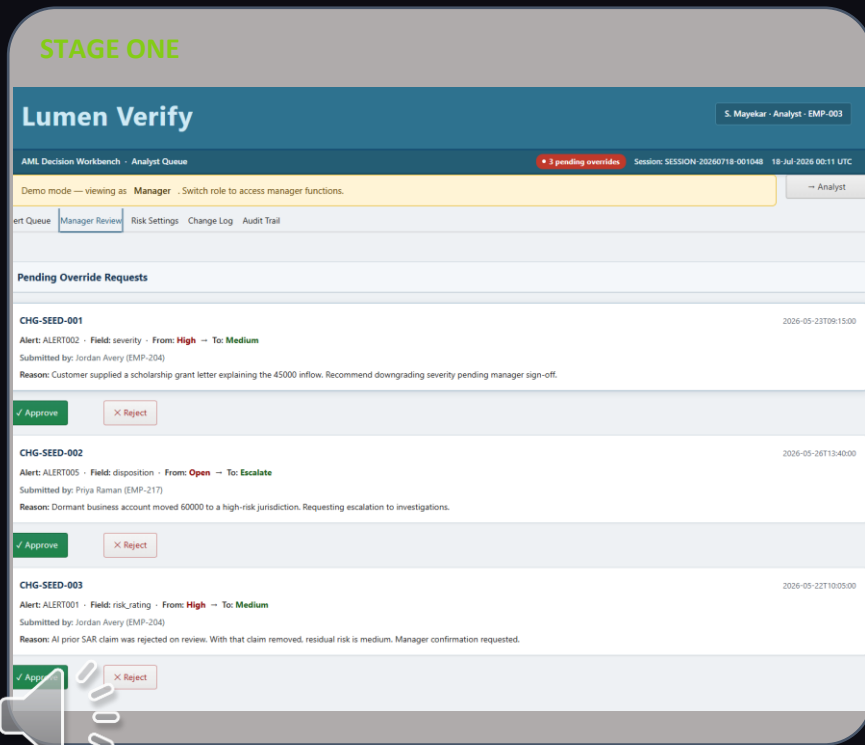
tests/ (765 tests)

Data integrity and demo cases
LLM drafter mock coverage
Verifier + negative paths
Demo seed governance

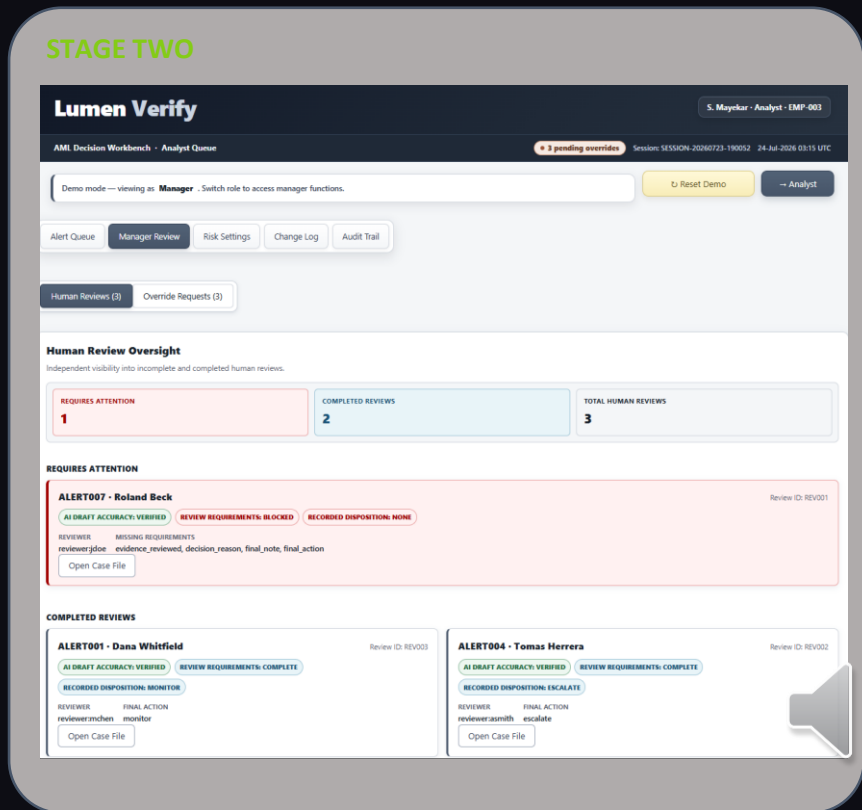


From Alert Queue to Decision Workbench

One screen, built in two stages.



Every commit is public in the linked repository.



The Lumen Control Stack

AI output is not trusted until it passes every control.

1

Closed Claim Vocabulary

AI can only emit 9 approved claim types. Anything outside is rejected.

2

Structured JSON Output

No free-form prose. Every claim is a typed, checkable data object.

3

Schema Enforcement

Tool schema + Python validation gates reject invalid AI output before verification.

4

Deterministic Verifier

Python checks each claim against source tables. PASS, FAIL, or NEEDS_REVIEW.

5

Human Approval Gate

A review is not complete until evidence, rationale, and a final note are recorded. Enforced in `review_gate.py`.

6

Controlled Audit Vocabulary

Every audit event must use an approved action name. Phantom events are caught by tests.

7

Examiner Replay

Examiner-reviewable audit trail reconstructs any decision with actor, action, timestamp, and approved event vocabulary.



We did not just add AI. We built seven controls around AI.

Hero Moment 1: The AI Gets Caught

Case File

Sylvie Marchand — R-VELOCITY

ALERT033 - CUST0027

AI DRAFT ACCURACY
UNSUPPORTED

REVIEW REQUIREMENTS
PENDING

RECORDED DISPOSITION
NONE

AI Draft Evidence Check

This checks whether the AI-generated draft matches the source evidence. It is not an AML clearance decision.

DRAFT CLAIM

rapid_movement = true

SOURCE EVIDENCE

no 24-hour window with 3+ in/out transactions summing over 10000

VERDICT

CONTRADICTED

DRAFT CLAIM

unusual_transaction_volume = true

SOURCE EVIDENCE

largest transaction 3896.00 does not exceed 2x expected_monthly_volume (10000.00)

VERDICT

CONTRADICTED

Missing source evidence: risk_assessment_memo

CUSTOMER PROFILE

Country	FR
Occupation	Retired
KYC Status	verified
KYC Current (12mo)	True
Prior SAR Count	0
Evidence Completeness	93%

TRANSACTIONS

TXN00155 - 2026-05-25T15:45:00	in 3415.0 (CA)
TXN00156 - 2026-05-26T18:09:00	out 3173.0 (JP)
TXN00157 - 2026-05-16T17:21:00	in 353.0 (FR)
TXN00158 - 2026-05-04T15:00:00	in 3656.0 (FR)
TXN00159 - 2026-05-24T18:09:00	out 1230.0 (AU)
TXN00160 - 2026-05-03T17:44:00	out 3896.0 (CA)

2 AI-drafted claims are contradicted by the source evidence. No completed human review is on file — the unsupported claim must not be accepted without human sign-off.

Human Review

Human review is required and awaiting submission.

Why This Wins

- The AI asserted two findings the source data does not support
- Deterministic Python caught it before the analyst ever saw it
- Verifier returns FAIL, not a warning, not a suggestion
- 32 of 93 AI claims across the dataset were contradicted by source evidence
- Every step is logged with actor and timestamp



Hero Moment 2: No Rubber-Stamping

Case File

Roland Beck — R-LARGE-WIRE

ALERT007 · CUST0007

AI DRAFT ACCURACY
VERIFIED

REVIEW REQUIREMENTS
BLOCKED

RECORDED DISPOSITION
NONE

AI Draft Evidence Check

This checks whether the AI-generated draft matches the source evidence. It is not an AML clearance decision.

DRAFT CLAIM

expected_activity_mismatch = true

SOURCE EVIDENCE

largest transaction 22000.00 exceeds expected_monthly_volume 10000.00

VERDICT

SUPPORTED

DRAFT CLAIM

prior_sar_history = true

SOURCE EVIDENCE

prior_sar_count=2 confirms prior SAR history for CUST0007

VERDICT

SUPPORTED

DRAFT CLAIM

missing_kyc_data = true

SOURCE EVIDENCE

at least one required evidence item is unavailable

VERDICT

SUPPORTED

Missing source evidence: risk_assessment_memo

CUSTOMER PROFILE

Country

CA

Occupation

Consultant

KYC Status

verified

KYC Current (12mo)

True

Prior SAR Count

2

Evidence Completeness

93%

TRANSACTIONS

TXN00014 · 2026-05-12T09:50:00

in 22000.0 (CA)

Seeded Demo Case: ALERT007, AI draft VERIFIED

REV001 (ALERT007) in human_reviews.csv has draft_disposition = accepted but decision_reason and final_note are empty and evidence_reviewed = false.

The gate detects this and blocks.

Current model-risk guidance reinforces the need for documented oversight, validation, and controls. This gate turns human accountability into a technical requirement, not a policy suggestion.

Human Review

REVIEWER	REVIEW DECISION	EVIDENCE REVIEWED
reviewerjdoe	accepted	False

DECISION RATIONALE
(none recorded)

FINAL NOTE
(none recorded)

HUMAN-REVIEW GATE: BLOCKED

Incomplete review detected. This stored review fails the same enforcement rule used by the decision pipeline. No final disposition is accepted.
Missing or invalid: evidence_reviewed, decision_reason, final_note, final_action

ENFORCEMENT: ENABLED

Hero Moment 3: The Happy Path

Structuring Alert. CUST0004. Two supported claims. Complete review, 100% evidence completeness.

1

AI Drafts

Ilm_drafter pulls transaction history and KYC profile. Emits structuring claim via JSON schema.

2

Verifier Confirms

Python checks: 2+ deposits in 72 hours, each between \$9,000 and \$9,999. Returns PASS.

3

Analyst Decides

Pre-assembled verified summary. SUPPORTED badges. Analyst reviews, writes justification, records the disposition.

45 min

Manual prep (ACAMS)

10 min

Lumen review (est.)

35 min

Time saved (est.)

3x

Throughput (est.)



Case File

Tomas Herrera — R-STRUCTURING

ALERT004 · CUST0004

AI DRAFT ACCURACY
VERIFIED

REVIEW REQUIREMENTS
COMPLETE

RECORDED DISPOSITION
ESCALATE

AI Draft Evidence Check

This checks whether the AI-generated draft matches the source evidence. It is not an AML clearance decision.

DRAFT CLAIM

structuring = true

SOURCE EVIDENCE

4 sub-threshold deposits within 72h consistent with structuring

VERDICT

SUPPORTED

DRAFT CLAIM

expected_activity_mismatch = true

SOURCE EVIDENCE

largest transaction 9700.00 exceeds expected_monthly_volume 8000.00

VERDICT

SUPPORTED

CUSTOMER PROFILE

Country	US
Occupation	Driver
KYC Status	verified
KYC Current (12mo)	True
Prior SAR Count	0
Evidence Completeness	100%

TRANSACTIONS

TXN00007 · 2026-05-14T12:00:00	in 9500.0 (US)
TXN00008 · 2026-05-15T12:00:00	in 9200.0 (US)
TXN00009 · 2026-05-15T12:00:00	in 9700.0 (US)
TXN00010 · 2026-05-16T12:00:00	in 9400.0 (US)

Human Review

REVIEWER

reviewer:asmith

REVIEW DECISION

edited

EVIDENCE REVIEWED

True

DECISION RATIONALE

Confirmed four sub-threshold deposits over three days; pattern consistent with structuring.

REVIEW REQUIREMENTS: COMPLETE

All required review fields are present. Review decision recorded: edited. Final case action: escalate.

ENFORCEMENT: ENABLED

Close

The Same Story, In the App

Hero Moment 3 in the actual Case File

Two **SUPPORTED** verdicts
AI Draft Evidence Check panel

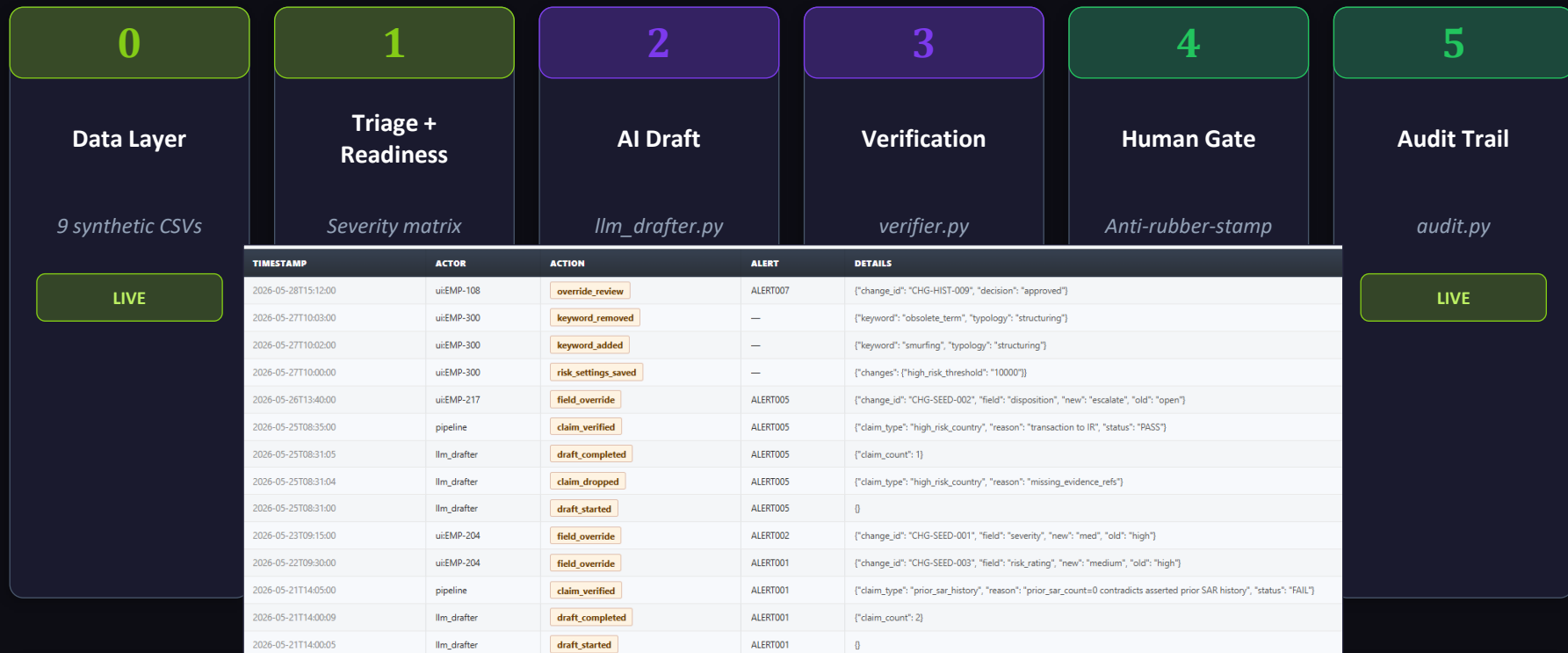
100% Evidence
Completeness
Customer Profile block

Complete human review
Rationale, evidence, and action

Gate opens
REVIEW REQUIREMENTS:
COMPLETE

System Architecture

Six architecture layers. Five run in the demo path. AI drafting is captured live and replayed deterministically.



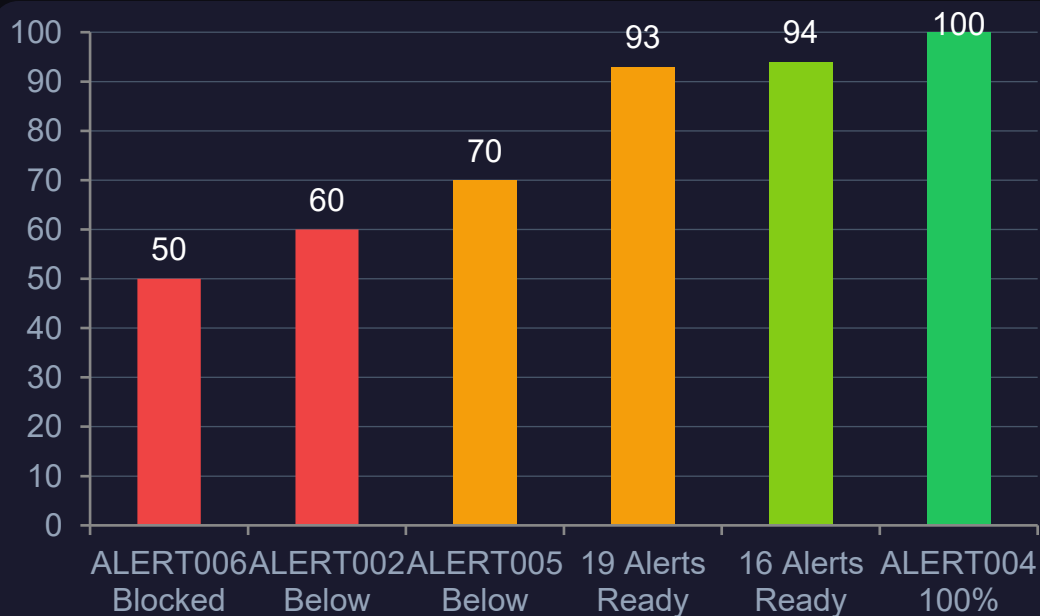
Streamlit UI · Surfaces all 6 layers to the analyst in a single workbench view



Evidence Completeness Engine

91.1% is not hardcoded. It is computed from source evidence.

Evidence Completeness = available evidence items / total required evidence items



50%

1 Blocked Case

ALERT006 only. Review gate blocks completion.

**60-
70%**

2 Below Threshold

ALERT002 at 60%, ALERT005 at 70%.
Review proceeds with caution.

**93-
94%**

36 Ready Cases

Pre-assembled by Lumen. Analyst can review.

91.1%

Overall Average

Evidence completeness computed from source data. Not hardcoded.



What Changes on Monday Morning

What the analyst faces	Manual review today	With Lumen
The analyst's hour	Most of it goes to gathering data from five systems.	Evidence arrives 91.1% assembled. The hour goes to judgment.
Catching AI mistakes	The analyst has to notice the error while reading.	Code flags the contradicted claim before the case is opened.
The examiner's file	Written up afterward, from memory, if at all.	A case cannot close without a rationale. The file writes itself.
Knowing where to start	Every alert looks equally reviewable until you dig in.	36 of 39 cases are review-ready. 1 is blocked on missing evidence.
When the AI is wrong	The wrong claim reaches the human and may be believed.	32 of 93 claims were contradicted. None reached an analyst.

The analyst stops collecting evidence and starts judging it. That change does not depend on the AI being good.



Built for Volume: What Happens at 10,000 Alerts

Cost to check one claim



1,000 alerts

10,000

100,000

Flat. Verification is code reading a table, with no AI call involved.

AI errors stopped



1,000 alerts

10,000

100,000

Rises with volume. Roughly one claim in three needs catching.

Only drafting scales in cost

and that is the piece a bank can throttle or route by severity.

Governance cannot loosen

a tenth claim type requires writing its check first, at any size.

More volume, more value

the control layer is worth more at scale, not less.

The checking never gets more expensive. The errors it stops keep growing. Volume works in your favor, not against you.



Business Value

35 min

saved per alert

Estimate. 45 min manual vs 10 min Lumen

3x

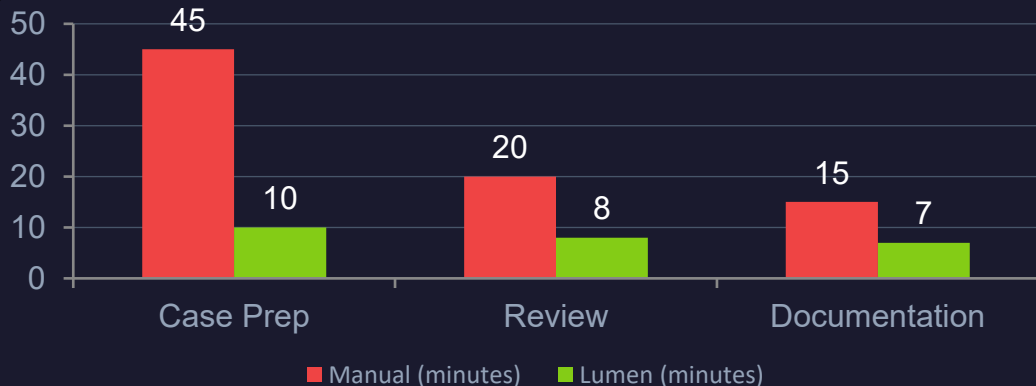
analyst throughput

Estimate. More low-level alerts cleared per day

\$45/hr

analyst loaded cost

BLS avg + 30% overhead



Sources

- *False-positive rate 90-95%: LexisNexis Risk Solutions*
- *Case prep 30-45 min: ACAMS benchmarks. Lumen-side times are design estimates, not measured.*
- *Analyst cost \$90K loaded: BLS + Glassdoor + 30% overhead*



Governance and Regulatory Alignment

SR 26-2

April 2026. Supersedes SR 11-7 and SR 21-8. Explicitly excludes generative AI from the legacy model risk framework. Lumen's verification spine is designed as a practical control layer around GenAI-assisted compliance work.

FinCEN Oct 2025 SAR FAQ

Supports disciplined SAR decisioning and value-over-volume reporting. Lumen preserves the evidence trail and human rationale behind AI-assisted case decisions.

FFIEC Examination

SAR narratives and supporting records must be accurate, complete, and explainable. Lumen preserves claim-level verification results and an examiner-reviewable audit trail.

If this system were deployed tomorrow at a global bank, would it align with regulator expectations?



How Lumen Differs From Common AI Compliance Tools

Category	Common AI-First Approach	Lumen Verify
AI Output	Natural-language summary	Structured claim object
Claim Scope	Open-ended model response	Closed 9-type vocabulary
Validation	Mostly prompt-based	Tool schema + Python gates
Verification	AI-assisted confidence	Deterministic source-table check
Failure Behavior	Analyst still sees weak output	Contradiction flagged before review
Human Review	Checkbox approval	Required rationale gate
Audit Trail	Activity log	Controlled audit vocabulary
Demo Reliability	Live model dependency	Seeded deterministic default
Regulatory Posture	Trust AI with review	Distrust AI until proven

Lumen does not make AI more autonomous. Lumen makes AI prove whether it deserves to be used.



Testing Is Part of the Governance Model

765 passing tests prove the system behaves as claimed.



Data Integrity

All 9 CSV tables exist and validate against Pydantic models. Core data is correct before any AI runs.



Closed Vocabulary

AI claims must match the approved 9-type vocabulary. Any claim outside it is rejected automatically.



Contradiction Detection

AI claims a pattern the source tables do not support. Verifier returns FAIL. Proven in test, not just data.



Review Gate

A review with no evidence check and no rationale evaluates as incomplete. The gate rule is tested and proven.



Verifier Behavior

Unknown claim types do not crash. Missing source returns NEEDS_REVIEW. Known claims route to real checks.



Demo Seed Governance

Pending overrides, audit history, and evidence readiness are generated deterministically. No surprises on demo day.



If this system were deployed tomorrow at a global bank

Would executives trust it?

Yes.

91.1% evidence completeness.
Documented audit trail. Model-risk-
control aligned.

Would it align with regulator
expectations?

Yes.

Aligned with SAR documentation
discipline. FFIEC examination-ready. Hard
human gate.

Would investigators use it?

Yes.

Pre-assembled, verified evidence instead
of a manual hunt across systems.

The model drafts. Code verifies. Humans decide.



References

Source backup for all statistics, regulatory citations, and industry benchmarks used in this presentation.

ROI + Industry Benchmarks

False-positive rate (90-95%)

LexisNexis Risk Solutions. True Cost of Financial Crime Compliance Study. Annual report. Also corroborated by Reuters AML industry reporting.

Manual case prep time (30-45 min)

ACAMS (Association of Certified Anti-Money Laundering Specialists). Industry benchmark data on manual triage process time across core banking systems.

Analyst loaded cost (\$90K / \$45/hr)

U.S. Bureau of Labor Statistics (BLS) occupational wage data for Compliance Officers + Glassdoor salary data for AML Analysts, with 30% standard corporate overhead applied.

Regulatory Framework

SR 26-2 (April 2026)

Federal Reserve Board. SR 26-2: Revised Guidance on Model Risk Management. April 2026. Supersedes SR 11-7 and SR 21-8. Excludes generative and agentic AI from scope.

FinCEN SAR FAQ (October 2025)

Financial Crimes Enforcement Network. Suspicious Activity Report (SAR) Filing FAQ. October 2025. Addresses SAR narrative completeness and value-over-volume reporting standards.

FFIEC SAR Examination Guidance

Federal Financial Institutions Examination Council. Bank Secrecy Act / Anti-Money Laundering Examination Manual. SAR narrative accuracy, completeness, and explainability standards.

Explore Lumen

See the code. Try the app. Read the finding.

LIVE INTERACTIVE SITE

leishka-pagan.github.io/Lumen

Architecture Six-layer control stack, click any layer.

Playground Run the deterministic verifier yourself.

The Finding The prompt-vs-code research result.

Scaling Cost model, 100 to 100,000 alerts.

SOURCE CODE

github.com/
[leishka-pagan/Lumen](https://github.com/leishka-pagan/Lumen)

765 tests · full audit history · public

CONNECT

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